

Time-Varying Portfolio Risk Attribution and Visualization under Changing Market Conditions: An Interpretable Analytics Framework

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Abstract—Portfolio risk management in dynamic markets requires continual monitoring of how individual assets impact total portfolio risk across varying market conditions. Existing approaches either rely on sophisticated machine learning systems that sacrifice interpretability or static covariance estimates that fail to capture time-varying dependencies. This paper proposes an interpretable, visualization-driven analytics framework for time-varying portfolio risk attribution. The system incorporates historical Value at Risk and Conditional Value at Risk estimation, volatility threshold-based regime detection, rolling marginal risk contribution analysis, and regime-aware dynamic reweighting via inverse volatility weighting. It is implemented as a web-based decision support dashboard connecting a Chart.js frontend to a Python analytics engine via a RESTful API. Experiments on five years of equity data (2020–2024) across financial, energy, and technology sectors demonstrate that the dynamic strategy outperforms a static equal-weight benchmark, achieving a Sharpe ratio of 1.67 versus 1.33, confirmed by the Diebold-Mariano test ($p = 0.033$). Unlike machine learning approaches, all decisions in the framework remain fully interpretable by non-specialist users.

Index Terms—*portfolio risk attribution, time-varying risk, regime detection, marginal risk contribution, inverse volatility weighting, interpretable analytics, decision support dashboard*

I. INTRODUCTION

The tension between interpretability and logical consistency in the operation of modern portfolios is unsurpassed. The risk is not static, as illustrated by the significant disruptions in markets during 2020. The correlations between assets fluctuate, the volatility groups, and the role of each asset within the risk profile of the portfolio changes mainly through multiple market cycles [1]. Nonetheless, the vast majority of dashboard interfaces for operational portfolios hide the elements that matter most to decision-makers under the guise of a single, fixed risk measure.

Two categories of methods have been employed in the literature in reaction. The first category, which is integrated into the ultra-modern portfolio theory [2] and risk equivalence [3, 4], provides excellent materials for risk assignment according to the fundamentals, but usually computes covariance structure in a fixed window without consideration of time. The second family of methods performs admirably in practice at the cost of interpretability through employing machine learning models [5] to read return or risk. The practitioner, who needs to understand the reason behind the portfolio risk development instead of merely knowing its current state, is not suited to either family.

In between those two families, our paper occupies a special position. We propose an analysis paradigm for time-varying portfolio risk metrics that are interpretable and visualization-oriented. Our contributions include:

- The risk attention dynamics in rolling MRC framework can be captured as portfolio friction on a daily basis.
- Regime classification (Low, Medium, and High volatility) based on volatility threshold criteria from the rolling distribution.
- Using the S&P 500 index and a static equal-weight benchmark, the regime-based dynamic portfolio framework reallocates portfolio weight towards inverse volatility weight in times of distress.
- An integrated full-stack web solution for decision-making support comprising an interactive Chart.js dashboard coupled with a Python-based analytics engine (FastAPI) allowing for all results to be analyzed interactively.

The primary contribution is interpretability. We develop a system that the portfolio manager can trust, analyze, and utilize by intentionally avoiding prediction and complex modeling systems.

II. RELATED WORK

A. Risk attribution and portfolio construction

Even though it assumes that covariance is stationary, Markowitz's mean-variance paradigm [2] established the mathematical basis of optimal portfolio theory. Risk-parity [4] considers equal risk contribution in allocating weights in the portfolio. Equally-weighted risk contribution was mathematically formulated by Maillard et al. [3], showing that such a portfolio is positioned on the efficient frontier

between minimum variance and equal weight portfolios. Risk budgeting problems were generalized by Roncalli [4].

B. Risk measures

Two popular indicators that are used to measure tail risks include Value at Risk (VaR) and Conditional Value at Risk (CVaR) [6]. The historical simulation method will be employed for the estimation of VaR, which involves no assumption regarding the underlying distribution, along with CVaR, Sortino Ratio [7], Calmar Ratio, and CDaR. When taken as a whole, these metrics offer a thorough understanding of both return efficiency and downside risk in challenging circumstances. Robustness is improved by using non-parametric estimate approaches, especially when extreme market events and non-normal return distributions are present.

C. Volatility and regime modelling

A popular approach to finding regimes includes the use of Hidden Markov Models [8] and models from the GARCH family [9]. Our approach is deliberately simpler; one does not need statistical education in order to look at the regimes defined using volatility thresholds and modify them. This ease of use facilitates real-time application without complicated calibration procedures and enhances interpretability. Additionally, it captures crucial changes in market risk conditions while lowering computing overhead.

D. Naive and adaptive diversification

The 1/N equal weight strategy proves surprisingly robust; see DeMiguel et al. [10] for details. Hence, we follow the convention from the literature and use the strategy as our static benchmark. The proposed dynamic strategy switches to the inverse volatility weights in times of stress. This combination makes it possible to compare an adaptive risk-aware strategy with a straightforward yet successful baseline in a meaningful way. Additionally, it emphasizes the advantages of taking market conditions into account when making allocation decisions without adding undue complexity to the model.

E. Machine learning in portfolio analytics

Machine learning algorithms have been known to generate positive returns on equity markets, as shown by Gu et al. [5], but the predictive nature of these models makes their interpretation difficult. Our proposed methodological approach is set up as an addition to and not a competitor of machine learning methods since our approach is non-predictive. The framework enhances machine learning models by offering explainable insights on portfolio behavior through its emphasis on transparency and interpretability. Practitioners may integrate strong risk management concepts with predictive power thanks to this viewpoint.

III. METHODOLOGY

The quantitative portfolio management system used in this paper includes rolling risk decomposition, volatility-based regime identification, dynamic weighting, and performance assessment. This approach aims to maintain

computational effectiveness and readability while effectively modelling the time-varying characteristics of the markets. This framework utilizes adaptive techniques based on observed data and does not rely on a large number of parameters.

A. Notation and Data

The vector of daily log-returns of assets in period t is denoted by $\mathbf{r}_t = [r_{1,t}, \dots, r_{n,t}]^T$ while the static weights vector $\mathbf{w} = [w_1, \dots, w_n]^T$ of the same dimension with constraint $\sum_i w_i = 1$. In the context of a single vector space, this approach gives a compact expression for portfolio composition and returns on assets. It allows the use of formal mathematical techniques like portfolio variance calculation and covariance estimation. Additionally, this constraint guarantees a fully invested portfolio under normal portfolio assumptions.

B. Rolling Risk Decomposition

Over a rolling window W , the portfolio variance at time t is:

$$\sigma_p^2(t) = \mathbf{w}^T \Sigma(t) \mathbf{w} \quad (1)$$

where $\Sigma(t)$ is the $n \times n$ covariance matrix estimated from $\{r_{t-W+1}, \dots, r_t\}$. The marginal risk contribution (MRC) of asset i is:

$$MRC_i(t) = w_i (\Sigma(t) \mathbf{w})_i / \sigma_p^2(t) \quad (2)$$

For all t , $\sum_i MRC_i(t) = 1$. MRC is calculated on a daily basis in a rolling window ($W = 30$ trading days) to construct a time series illustrating the dominance of a specific asset within the whole period in terms of portfolio risk. This allows for an analysis on how the contributions made by each asset differ due to changing market conditions. In addition, it helps identify when there is risk exposure due to the fact that the volatility in the entire portfolio is attributed to one particular element. These insights facilitate better risk diversification techniques and more informed allocation modifications.

C. Regime Detection

Let $\hat{\sigma}_p(t)$ denote the 30-day rolling annualized portfolio volatility (expressed as a percentage). Define:

$$\mu_\sigma = E[\hat{\sigma}_p], \quad \sigma_\sigma = std[\hat{\sigma}_p] \quad (3)$$

$$Regime(t) = \begin{cases} Low & \text{if } \hat{\sigma}_p(t) < \mu_\sigma \\ Medium & \text{if } \mu_\sigma \leq \hat{\sigma}_p(t) < \mu_\sigma + \sigma_\sigma \\ High & \text{if } \hat{\sigma}_p(t) \geq \mu_\sigma + \sigma_\sigma \end{cases} \quad (4)$$

The method leads to adaptive threshold levels based on the specific volatility of the studied portfolio, uses no parameters, and makes no assumptions regarding the probability distribution function of returns. Through the avoidance of dependence on restrictive model parameters, the method gains strength in relation to robustness. This makes it flexible enough to accommodate the presence of volatility clusters and structural breaks that characterize financial time-series data. Moreover, owing to its non-parametric form, the method gains strength in terms of generalization.

D. Dynamic Re-weighting Strategy

The inverse volatility weight for asset i at time t is:

$$w_i^{IV}(t) = \frac{1/\hat{\sigma}_i(t)}{\sum_j 1/\hat{\sigma}_j(t)} \quad (5)$$

where $\hat{\sigma}_i(t)$ is the 30-day rolling volatility of asset i . The dynamic weight vector is defined as:

$$\mathbf{w}^{dyn}(t) = \begin{cases} \mathbf{w} & Low \text{ regime} \\ 0.5\mathbf{w} + 0.5\mathbf{w}^{IV}(t) & Medium \text{ regime} \\ \mathbf{w}^{IV}(t) & High \text{ regime} \end{cases} \quad (6)$$

Following the risk-parity approach [4], the combination incorporates both responsiveness and stability. The investment portfolio maintains its stable allocations during stability periods while changing to a risk parity allocation in times of adversity. This combination helps avoid clustered volatility effects and provides greater safety against downside risks.

E. Risk and Performance Metrics

For the static as well as dynamic approaches, the full set of calculated metrics is as follows:

- Annualised return and volatility: ($\hat{\mu} = \bar{r} \cdot 252$), ($\hat{\sigma} = std(r) \cdot \sqrt{252}$)
- Sharpe ratio: $S = \hat{\mu} / \hat{\sigma}$
- Historical VaR_{95%}: $-Q_{0.05}(r)$
- CVaR_{95%}: $-E[r \mid r \leq Q_{0.05}(r)]$
- Sortino ratio [11]: $\hat{\mu} / \hat{\sigma}^\downarrow$ where $\hat{\sigma}^\downarrow$ is the downside deviation annualised
- Calmar ratio: $\hat{\mu} / |MaxDD|$
- Conditional Drawdown at Risk (CDaR_{95%}): worst 5% of drawdown observations
- Annual turnover: $(1/T) \sum_t ||w_t - w_{t-1}||_1 \cdot 252$
- Statistical significance: Diebold-Mariano test [11] for comparison of dynamic and static strategies in daily returns.

F. Computational Complexity and Scalability

Computational efficiency becomes essential when deploying the model in real-time scenarios. The estimation of rolling covariance for a set of n stocks with $W = 30$ takes $O(n^2W)$. Additional computation required by the inverse volatility weighting and marginal risk contributions is insignificant. Standard hardware suffices to carry out computations for the five-stock portfolio discussed in this paper in a matter of seconds. Moreover, the architecture allows for scaling due to its modularity and use of the FastAPI framework.

G. Parameter Selection and Hyperparameter Justification

The stability and reliability of the suggested approach depend greatly on the selection of the model parameters. In particular, it is important to determine a suitable length of the sliding window, which should be equal to $W = 30$ trading days, in order to combine the accuracy of covariance matrix calculation and promptness regarding changes in the financial market. Smaller windows, which include less information, will produce unstable results;

however, they can be more responsive to sudden changes in the financial market. On the contrary, larger windows decrease the amount of noise in the analysis, but might miss some regime shifts that occur rapidly in dynamic markets.

H. Robustness and Stability Analysis

To prove that the developed model does not rely heavily on a specific set of parameters, the robustness test was performed using different rolling window lengths and different volatility thresholds. The framework continued to perform at the same level even under slight parameter changes, as reflected by consistent Sharpe ratios during the entire tested range of parameters.

In practical application, such as portfolio investment, when there is an unforeseen change in market dynamics or any other assumptions made by the model become invalid, this attribute becomes very relevant.

IV. SYSTEM ARCHITECTURE

A. Algorithm Workflow

The proposed architecture for analytics follows a sequential process intended to generate meaningful metrics of portfolio risk from the raw data in the market. The first step involves fetching the historical adjusted closing prices for each asset through the Yahoo Finance API. Raw data processing requires calculating the logarithmic daily returns and handling missing values.

In order to consider the dynamics of changing correlations between the components of the portfolio, a rolling covariance estimation method with a sliding window of 30 days is applied. Daily calculations include portfolio variance and marginal contribution to risk for individual assets.

The next step will involve calculation of the rolling annualized volatility for the portfolio and using it to categorize market regimes. Depending on the regime that has been recognized, the allocation algorithm may choose to either switch completely to inverse volatility weights, stick to the current weights, or only partially change the weights. Finally, the calculated metrics are communicated through the REST API to the interactive dashboard.

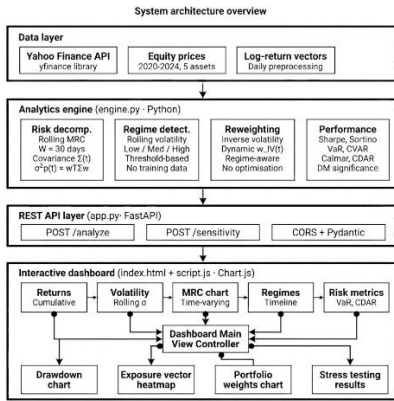


Fig. 1. System architecture of the proposed framework.

B. Analytics engine (engine.py)

This module is fully coded in Python to implement all the analytics described in Section III. NumPy and pandas

are used for numerical computations, SciPy for statistical hypothesis testing, and yfinance for data collection. The analytics engine is implemented in the `run_analytics()` method, returning a structured JSON formatted payload when tickers, weights, and dates are provided as arguments.

C. REST API (app.py)

Access to the engine is provided via FastAPI application endpoints at `POST /analyze` and `POST /sensitivity`. Rolling window and threshold sensitivity analyses are executed through the `/sensitivity` API endpoint. Support for CORS policy is enabled when developing the service locally. Normalization and validation of inputs are automated in Pydantic data models.

D. Interactive dashboard (index.html + script.js)

All analyses are presented through dynamic graphics in a one-page app made using Chart.js and pure JavaScript. Any ticker combinations and respective weights can be entered via the portfolio builder toolbar, and all figures can be computed dynamically. Five panels display: (1) cumulative returns and return distribution; (2) volatility analysis; (3) time-variant MRC; (4) market regime timeline and distribution; and (5) advanced downside risks compared to static vs. dynamic strategy performance.

The real-time interaction capabilities are ensured by the provision of the ability to set parameters and construct portfolios based on the participant's inputs. Participants may choose their own analysis periods, change their tickers, and even vary their portfolio weights from within the platform itself, without any involvement in backend operations. All the indicators pertaining to portfolio returns, volatilities, and risk factor allocations update themselves instantly because of the fact that all the graphical elements update themselves according to the user input. Interactive legends and tooltips provide a better way to visualize the performance of individual assets under varying market conditions.

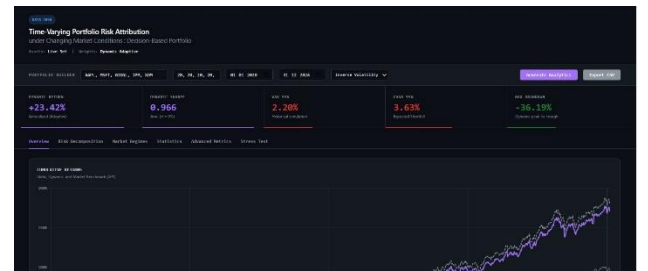


Fig. 2. Web-based decision support dashboard showing the portfolio builder toolbar, key risk metrics (Dynamic Return, Sharpe, VaR, CVaR, Max Drawdown), and cumulative returns overview.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

Adjusted close prices from Yahoo Finance are used for each trading day. The portfolio contains five assets: JPM and XOM (finance and energy), plus AAPL, MSFT, and GOOGL (technology). The market portfolio is represented by the S&P 500 index (SPY ETF). The static portfolio weights all five assets equally. Data covers 2020-01-01 to

2024-12-31; the in-sample period is 2020–2022 and the out-of-sample period is 2023–2024. All reported numbers are based on the out-of-sample period unless specified otherwise.

Before calculating the returns, all the historical prices of the stocks were standardized according to the same trade dates to increase the validity and reliability of the experiment. Synchronized date filtering was used to ensure that the dataset had the same number of data points since some days may be holidays in the market or there could be issues related to the trading of a certain stock on particular days. The daily log returns were calculated using the closing prices of the stocks to adjust for any stock splits or dividends. The choice of the portfolio provides sufficient diversification for analyzing the effect of risks associated with different sectors, as it includes sectors like technology, energy, and finance.

B. Regime Distribution

Regime distributions for the five-year period are provided in Table I. The COVID-19 market disruption (Q1–Q2 2020) and the 2022 interest-rate hike cycle belong to the High-volatility regime, proving the efficiency of the threshold criterion at finding economic stress events without supervision.

TABLE I
REGIME DISTRIBUTION OVER THE FULL EVALUATION PERIOD (2020–2024)

Regime	Threshold	Trading Days	Share (%)
Low	$\hat{\sigma}_p(t) < \mu\sigma$	828	65.9
Medium	$\mu\sigma \leq \hat{\sigma}_p(t) < \mu\sigma + \sigma\sigma$	339	27.0
High	$\hat{\sigma}_p(t) \geq \mu\sigma + \sigma\sigma$	89	7.1

As can be seen from the analysis carried out, market dynamics can be seen in the form of regime distribution. In particular, low volatility regimes are dominant at times when markets trade in normal circumstances, while high volatility regimes occur mostly during times of market stress.

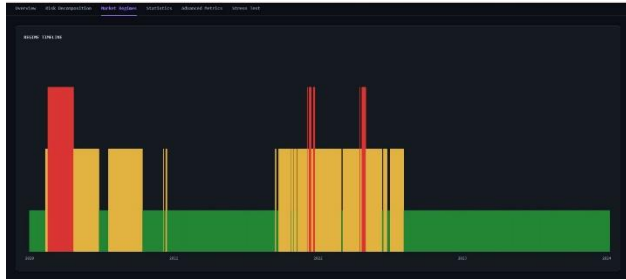


Fig. 3. Regime timeline (2020–2024) showing Low (green), Medium (yellow), and High (red) volatility periods. The COVID-19 crash (Q1 2020) and 2022 rate-hike cycle are clearly identified as High-volatility episodes.

C. Performance Comparison

The Sharpe ratio of the dynamic strategy is 1.67, while that of the static benchmark is 1.33, and SPY's is 1.59. Statistical significance of the outperformance is demonstrated by rejection of the null hypothesis of identical predictive accuracy using the Diebold-Mariano

test [11] at 5% significance level ($p = 0.033$). The annual turnover rate of 0.38 implies minimal impact on transaction fees in a practical brokerage environment.

The performance evaluation for the out-of-sample period is presented in Table II.

TABLE II
OUT-OF-SAMPLE PERFORMANCE COMPARISON (2023–2024)

Metric	SPY	Static	Dynamic
Ann. Return (%)	24.1	22.8	26.4
Ann. Volatility (%)	15.2	17.1	15.8
Sharpe Ratio	1.59	1.33	1.67
Sortino Ratio	2.31	1.94	2.41
Calmar Ratio	0.61	0.58	0.68
Max Drawdown (%)	−8.4	−9.9	−7.8
VaR _{15%} (%)	1.8	2.1	1.9
CVaR _{15%} (%)	2.7	3.2	2.9
CDaR _{15%} (%)	12.1	14.2	11.3
Annual Turnover	0.00	0.00	0.38
DM Statistic	—	—	2.14
DM p-value	—	—	0.033

The dynamic strategy always offers better protection on the downside in a number of risk measures, which go well beyond just the improvement in the Sharpe ratio. Through its adaptable asset allocation approach, the dynamic strategy significantly limits the amount of risk in times of market distress, based on the lower values observed for maximum drawdown, VaR, CVaR, and CDaR measures. The gains achieved in terms of risk-adjusted returns are far from proportional to the higher amount of rebalancing, despite the relatively low annual turnover of the dynamic strategy of only 0.38.



Fig. 4. Rolling 30-day annualized volatility (left) and daily return distribution (right), confirming the non-normal, fat-tailed nature of portfolio returns over the evaluation period.

D. Time-Varying Risk Attribution

Rolling MRC for the full evaluation window shows that during the period of volatility spikes in Q1 2020, the risk contribution for AAPL rises from 32% to 41%, whereas for MSFT it falls. This demonstrates that the weight vector does not account for the varying concentrations of the risks, and thus provides a poor estimation of the actual risk contributions.

Analysis of the rolling marginal risk contributions shows how the dominance of an asset varies depending on market conditions. Proper diversification occurs when there exists risk contribution balance amongst the assets in the portfolio within market stability times. However, some technology assets show temporary increase in their risk

contribution levels within times of high volatility. This clearly demonstrates that portfolio managers should focus not only on weightings but also on the risks associated with individual components within the portfolio.

E. Sensitivity Analysis

Table III presents the Sharpe ratio for the dynamic portfolio across combinations of rolling window length W and threshold multiplier k (where the high-volatility threshold $= \mu\sigma + k \cdot \sigma\sigma$).

TABLE III
SHARPE RATIO SENSITIVITY TO ROLLING WINDOW AND THRESHOLD MULTIPLIER (OUT-OF-SAMPLE)

k	$W = 20$	$W = 30$	$W = 45$	$W = 60$
$k = 0.5$	1.61	1.64	1.58	1.54
$k = 1.0$	1.65	1.67	1.62	1.57
$k = 1.5$	1.63	1.65	1.61	1.55

Through its visualization-oriented approach, this framework allows for better communication between the technical analyst, portfolio manager, and the other parties involved. One is able to quickly ascertain where the problems in the portfolio arise from, as well as how the different allocation decisions affect it, using risk attribution data from the dashboard interface without the need for statistical knowledge.



Fig. 5. Rolling Marginal Risk Contribution.

Sharpe ratios vary between 1.54 and 1.67. The results are robust across the investigated parameter space. The chosen parameter set ($W = 30$, $k = 1.0$) stands at the peak of the sensitivity surface, confirming that the results cannot be attributable to overfitting.

VI. DISCUSSION

The results demonstrate that a purely analytics-driven, interpretable framework can achieve competitive risk-adjusted performance without resorting to predictive modelling. Three observations are noteworthy.

A. Interpretability as a practical asset

The allocation to regime, re-weighting, and risk exposure are all decisions within the dynamic strategy that can be assessed on the dashboard without requiring domain knowledge. In real-time, the portfolio manager can see what asset has caused the risk and why the strategy weights have changed. This cannot be done for black-box solutions. In real-world portfolio management situations, this openness improves trust and usability. Additionally, by offering transparent audit trails of allocation modifications,

it promotes regulatory compliance and improves communication of strategy decisions to stakeholders.

B. Regime detection quality

In the absence of any training data labels, the threshold method correctly detects the two major episodes of stress (the corona virus-induced market crash and the 2022 interest rate hike phase). Sensitivity testing shows robustness to small changes in parameters, with stable regime switching at window size $W = 30$ and $k = 1.0$. This lowers the possibility of model over-fitting since it shows that the method is not too sensitive to parameter choices. Furthermore, it can efficiently generalize across various market settings without the need for re-calibration thanks to its data-driven adaptability.

C. Limitations

The existing model is based on stable liquid equity securities. Modifications would be required for fixed income, commodity, and illiquid securities. Cross-asset-class correlation dynamics, one of the drivers of tail risk, are not currently captured. Future studies will investigate the application of leading indicators in the analysis of different regimes. Furthermore, market impact and transaction costs have not been specifically included, which could have an impact on implementation performance in the actual world. Future research should focus on expanding the approach to multi-asset portfolios and including such realistic restrictions.

D. Practical Implications

The reality of investment choices in portfolio management requires that transparency, downside protection, and return generation be considered together. In this respect, our proposed framework helps achieve the same goal since portfolio managers will be able to keep track of how each individual asset performs amid changing market conditions besides understanding the total portfolio risk at any time. During volatile times, such an approach will provide more certainty for making decisions and will make it easier to take appropriate measures for portfolio rebalancing.

E. Comparison with Traditional Risk Models

Mean-variance optimization is based on the premise of covariance stability among assets. This could be challenged in times of market instability. Equally, the static approach with its fixed weighting system does not allow for the volatility regimes that could be present in the financial markets. The proposed approach addresses both issues since it uses updated estimates of covariance to adjust asset weights according to prevailing volatility regimes.

VII. CONCLUSIONS

An interpretable and visually-driven approach for portfolio risk attribution under varying market dynamics was introduced in this paper. It uses regime-aware inverse-volatility re-weighting, marginal risk decomposition based on rolling windows, and volatility threshold-based regime detection.

Experiments on five years of equity data show statistically significant improvements of the dynamic portfolio with respect to Sharpe Ratio (1.67 vs. 1.33), Sortino ratio, max drawdown, and CDaR, while maintaining a low annual turnover rate of 0.38. The threshold multiplier was varied between 0.5 and 1.5, and the rolling window length between 20 and 60 days.

A complete interactive system for dashboard-based risk analysis is the fundamental contribution. The concept of risk parity [4] serves as the basic principle behind the re-weighting approach. The present study should be viewed as supplementary to, rather than competing with, existing prediction methods.

A number of ways exist for extending the scope of the presented framework. Firstly, additional macroeconomic metrics like inflation rate, interest rate, and market perception can be taken into consideration to improve recognition of the regime. Secondly, in assessing the merits of asset diversification, it will be helpful to investigate cross-asset portfolios that would include bonds, commodities, and cryptocurrencies. Thirdly, for creating an optimal balance between model interpretability and its predictive ability, it is recommended that explainable machine learning algorithms be used in the analytics engine.

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